PASSING PROPERTIES OF A DECENT SPACE OVER A BASE TO PROPERTIES OVER A PROPERTY-DETERMINED POINT

THE STACKS PROJECT AUTHORS

Editorial note. The base-change boundedness input is not carded and is declared external.

Allowed standard prerequisites (editorial).

- Base change of universal boundedness. lemma-base-change-universally-bounded (same chapter, NOT carded, declared external). Boundedness of the fibres over the base point.
- Residual space factorization. lemma-factor-through-residual-space-decent (carded L26728). Factorization through the residual space at the base point.

Source reference. Original lemma 03JN: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 327–327. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JM: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 297–297. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HF: focused-seeds/geometry/S9-decent-property-over-property/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HG: focused-seeds/geometry/S9-decent-property-over-property/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $\omega \in \{\beta, decent, reasonable\}$. Suppose that Y has property (ω) and $f : X \to Y$ has (ω). Then X has (ω).*

Proof. Let us prove the lemma in case $\omega = \beta$. In this case we have to show that any $x \in |X|$ is represented by a monomorphism from the spectrum of a field into X. Let $y = f(x) \in |Y|$. By assumption there exists a field k and a monomorphism

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$\mathrm{Spec}(k) \to Y$ representing y. Then x corresponds to a point x' of $\mathrm{Spec}(k) \times_Y X$. By assumption x' is represented by a monomorphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k) \times_Y X$. Clearly the composition $\mathrm{Spec}(k') \to X$ is a monomorphism representing x.

Let us prove the lemma in case $\omega = \textit{decent}$. Let $x \in |X|$ and $y = f(x) \in |Y|$. By the result of the preceding paragraph we can choose a diagram

$$\begin{array}{ccc} \mathrm{Spec}(k') & \xrightarrow{x} & X \\ \downarrow & & \downarrow f \\ \mathrm{Spec}(k) & \xrightarrow{y} & Y \end{array}$$

whose horizontal arrows monomorphisms. As Y is decent the morphism y is quasi-compact. As f is decent the algebraic space $\mathrm{Spec}(k) \times_Y X$ is decent. Hence the monomorphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k) \times_Y X$ is quasi-compact. Then the monomorphism $x : \mathrm{Spec}(k') \to X$ is quasi-compact as a composition of quasi-compact morphisms (use Morphisms of Spaces, Lemmas 03HF and 03HG). As the point x was arbitrary this implies X is decent.

Let us prove the lemma in case $\omega = \textit{reasonable}$. Choose $V \to Y$ étale with V an affine scheme. Choose $U \to V \times_Y X$ étale with U an affine scheme. By assumption $V \to Y$ has universally bounded fibres. By Lemma 03JN the morphism $V \times_Y X \to X$ has universally bounded fibres. By assumption on f we see that $U \to V \times_Y X$ has universally bounded fibres. By Lemma 03JM the composition $U \to X$ has universally bounded fibres. Hence there exists sufficiently many étale morphisms $U \to X$ from schemes with universally bounded fibres, and we conclude that X is reasonable. $\square$